Divyanshu Bansal

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TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, C++, Java
Backend : Node.js, Express.js, Hono, Bun, FastAPI, Django REST Framework, WebSockets, REST APIs
LLMs & Generative AI: LLM architecture (RoPE, GQA, SwiGLU), RAG pipelines, fine-tuning, agent frameworks,n8n
Frontend: Next.js 14 (App Router), React, Tailwind CSS, Framer Motion
Databases: PostgreSQL (Neon), MongoDB, Supabase, Prisma ORM, Redis
Cloud / DevOps: Vercel, Cloudinary, Docker, Git, GitHub, Postman, CI/CD
ML / Data: TensorFlow, scikit-learn, pandas, NumPy, Matplotlib, Random Forest, Remote Sensing,GIS

EXPERIENCE

Research & Development Intern June 2024 – July 2024
Benciti Technology Ltd. *Bengaluru, India*

- **Built and integrated REST APIs** across backend systems, coordinating with senior engineers to resolve **database performance bottlenecks** and ensure stable service behavior under production load.
- **Contributed** to backend testing pipelines and UI/UX debugging for the **Nudge app**, identifying and fixing edge cases across API response flows and validation layers.
- **Participated in Agile sprint cycles**, communicating blockers proactively, reviewing peer code, and iterating on solutions based on QA and team feedback.

PROJECTS

ClarityAI | *Next.js 14, TypeScript, Gemini API, Framer Motion, Vercel* 2026

- **Architected and deployed** an AI-powered text processing app at **clarityai-olive.vercel.app** that converts unstructured input into **four structured outputs** — TL;DR, Action Items, Key Decisions, and Open Questions — via **Gemini API** with automatic fallback to gemini-flash-lite on quota exhaustion.
- **Engineered** a streaming follow-up chat interface with token-by-token responses, conversation context retention, and rate-limit-aware request handling for iterative Q&A on processed content.
- **Implemented** server-side API key security via **Next.js Route Handlers**, error boundaries for graceful failure states, and **Vercel edge deployment** achieving **sub-100ms cold starts**.

SnapCut AI | *React, Vite, Supabase, Cloudinary, ClipDrop API, n8n, Razorpay* 2026

- **Built and deployed** an end-to-end **AI background-removal SaaS** at **snapcut-ai-woad.vercel.app**: upload → **Cloudinary** temp storage → **n8n webhook pipeline** → **ClipDrop API** processes image → result returned to frontend with **before/after slider preview**.
- **Designed an 8-node n8n automation workflow** (Webhook → Validate → Fetch → AI Process → Upload → Supabase Insert + Credit Deduction → Respond) replacing a custom backend server entirely and **reducing infrastructure cost to near zero**.
- **Integrated Supabase Auth** (email/password + Google OAuth) with PostgreSQL credit tracking via a stored procedure, **Razorpay Orders API** with **HMAC webhook verification**, and 24-hour auto-delete Cloudinary storage; deployed to **Vercel** from github.com/audicitty/snapcutAI.

Spatio-Temporal Urban Expansion — Gurgaon (B.Tech Thesis) | *Python, Scikit-learn, Landsat-8, Sentinel-1 SAR, Sentinel-2, VIIRS NTL* 2024 – 2026

- **Built** an end-to-end **multi-phase geospatial ML pipeline** analyzing both **horizontal sprawl and vertical densification** of Gurgaon across **2015, 2020, and 2023** using **Landsat-8**, **Sentinel-2** optical, **Sentinel-1 SAR**, and **VIIRS** nighttime-lights data.
- **Trained a Stratified Random Forest** 6-class land-cover classifier achieving **92%+ accuracy** across **24 images**; engineered spectral indices (**NDVI, NDBI, MNDWI**) and **SAR backscatter** thresholds as discriminative features for built-up vs. peri-urban zones.
- **Designed a vertical-growth detection module** combining **Sentinel-1 SAR backscatter** intensity with **VIIRS nighttime-light** density as a high-rise densification proxy; integrated a **Neutrosophic (T-I-F) uncertainty framework** to flag low-confidence boundary pixels.

ACHIEVEMENTS & CERTIFICATIONS

- **500+ DSA problems solved** across different coding platforms · **Certifications:** AI for Everyone — DeepLearning.AI (Coursera ID: QMZVLLGSTJ4X), Internet of Things — IIT Bombay, Mastering Data Structures & Algorithms.

EDUCATION

BML Munjal University, Gurugram July 2022 – May 2026
B.Tech, Computer Science & Engineering — CGPA: 7.82 / 10